Define a feature vector in Multimedia IR and give one image feature example (2 marks)

A **feature vector** is a numerical representation of an object's visual, textual, or audio properties, used by multimedia IR systems to index, compare, and retrieve similar items.

- **Definition:** An ordered set of extracted feature values, e.g., (\mathbf{v} = (f_1, f_2, \dots, f_n)), that describes an object in a high-dimensional feature space.
- **Purpose:** Enables similarity computation (e.g., Euclidean or cosine distance) between a query and stored objects.

Example of an image feature: Color histogram — a vector counting the frequency of pixels in each color bin, e.g., ((0.35, 0.20, 0.15, \dots)).

Result: A feature vector is a numeric descriptor of an object; a color histogram is one common image feature vector.

Differentiate between browsing and searching on the Web (2 marks)

Aspect	Browsing	Searching
Goal	Exploratory, open-ended navigation	Targeted retrieval of a specific answer
User input	Follows links, categories, or menus	Enters keywords/query into a search box
Outcome	Serendipitous discovery	Ranked list of relevant results

Key Distinctions: Browsing is user-driven navigation without a precise target, whereas searching is query-driven retrieval aimed at a known information need.

Define starting points in a search interface and give two examples (2 marks)

Starting points are entry mechanisms or seed resources offered by a search interface that help users begin their information-seeking task without needing to formulate a full query.

- **Examples:**
  1. **Directory/category links** — e.g., Yahoo!-style subject hierarchies.
  2. **Suggested queries or hot topics** — e.g., trending searches on Google.

Result: Starting points lower the cognitive barrier to initiating a search and guide users toward productive queries.

Define SeeSoft and state its use (2 marks)

SeeSoft is a **visualization tool** that displays large text files (such as source code) as colored line-based views, where each line is represented as a thin colored strip.

- **Use:** It helps users perceive patterns, anomalies, and structure in large documents at a glance.
- **Application:** Software maintenance, code analysis, and document visualization in IR interfaces.

Result: SeeSoft provides a compact visual overview of large text collections to aid navigation and pattern detection.

Define metadata and list two types (2 marks)

Metadata is **data about data** — structured information that describes, explains, locates, or otherwise makes it easier to retrieve and manage an information resource.

- **Two types:**
  1. **Descriptive metadata** — e.g., title, author, subject keywords.
  2. **Structural metadata** — e.g., chapter hierarchy, page order, file format details.

Result: Metadata improves resource discovery, organization, and interoperability across IR systems.

Define a meta-search engine and give one example (2 marks)

A **meta-search engine** is a system that sends a user's query to **multiple independent search engines simultaneously**, aggregates their results, and presents a unified ranked list.

- **Characteristics:** No own index; relies on other engines' indexes; removes duplicates and re-ranks.
- **Example: Dogpile** (also MetaCrawler, Searx).

Result: Meta-search engines increase coverage and reduce dependence on a single engine's ranking.

Define stemming and explain with two examples (2 marks)

Stemming is the process of reducing inflected or derived words to their **root or stem form**, so that morphologically related terms match during retrieval.

- **Example 1:** "retrieval", "retrieve", "retrieved" → **retriev**
- **Example 2:** "studies", "studying", "studied" → **studi**

Result: Stemming improves recall by conflating word variants into a single index term.

Define HCI and state its importance in an IR user interface (2 marks)

Human-Computer Interaction (HCI) is the study of how people interact with computer systems, focusing on the design, evaluation, and implementation of usable interfaces.

- **Importance in IR:**
  - Helps design intuitive query formulation and result presentation.
  - Supports relevance feedback, query refinement, and result visualization.
  - Reduces cognitive load and improves user satisfaction and search effectiveness.

Result: HCI principles are essential for making IR interfaces usable, efficient, and user-centered.

Define KWIC and state where it is used (2 marks)

KWIC (Key Word In Context) is a display format in which each occurrence of a search term is shown **within its surrounding text context**, typically aligned in a central column.

- **Format example:**
  - ...information **retrieval** is the study...
  - ...web search is a type of **retrieval**...
- **Uses:** Concordances, digital libraries, legal document search, and result snippet presentation.

Result: KWIC helps users quickly judge the relevance of a term by seeing its contextual usage.

Differentiate between HTML and XML (2 marks)

Aspect	HTML	XML
Purpose	Presentation and display of web content	Description, storage, and transport of structured data
Tags	Predefined, fixed set	User-defined, extensible
Case sensitivity	Case-insensitive	Case-sensitive
Closing tags	Some optional	Mandatory for all elements

Key Distinctions: HTML focuses on how data looks; XML focuses on what data means, making it suitable for data interchange.

Explain suffix trees and suffix arrays with an example; search 'ana' in 'banana' (5 marks)

Suffix Tree: A compressed trie containing all suffixes of a string, where each edge is labeled with a substring and each leaf represents a suffix position. It supports fast pattern matching in (O(m)) time for pattern length (m).

Suffix Array: A sorted array of all suffix starting positions of a string. It is more space-efficient than a suffix tree and supports binary-search-based pattern matching in (O(m \log n)).

Example — Text: `banana` (positions 0–5)

Suffixes:

- 0: banana
- 1: anana
- 2: nana
- 3: ana
- 4: na
- 5: a

Sorted suffixes and suffix array:

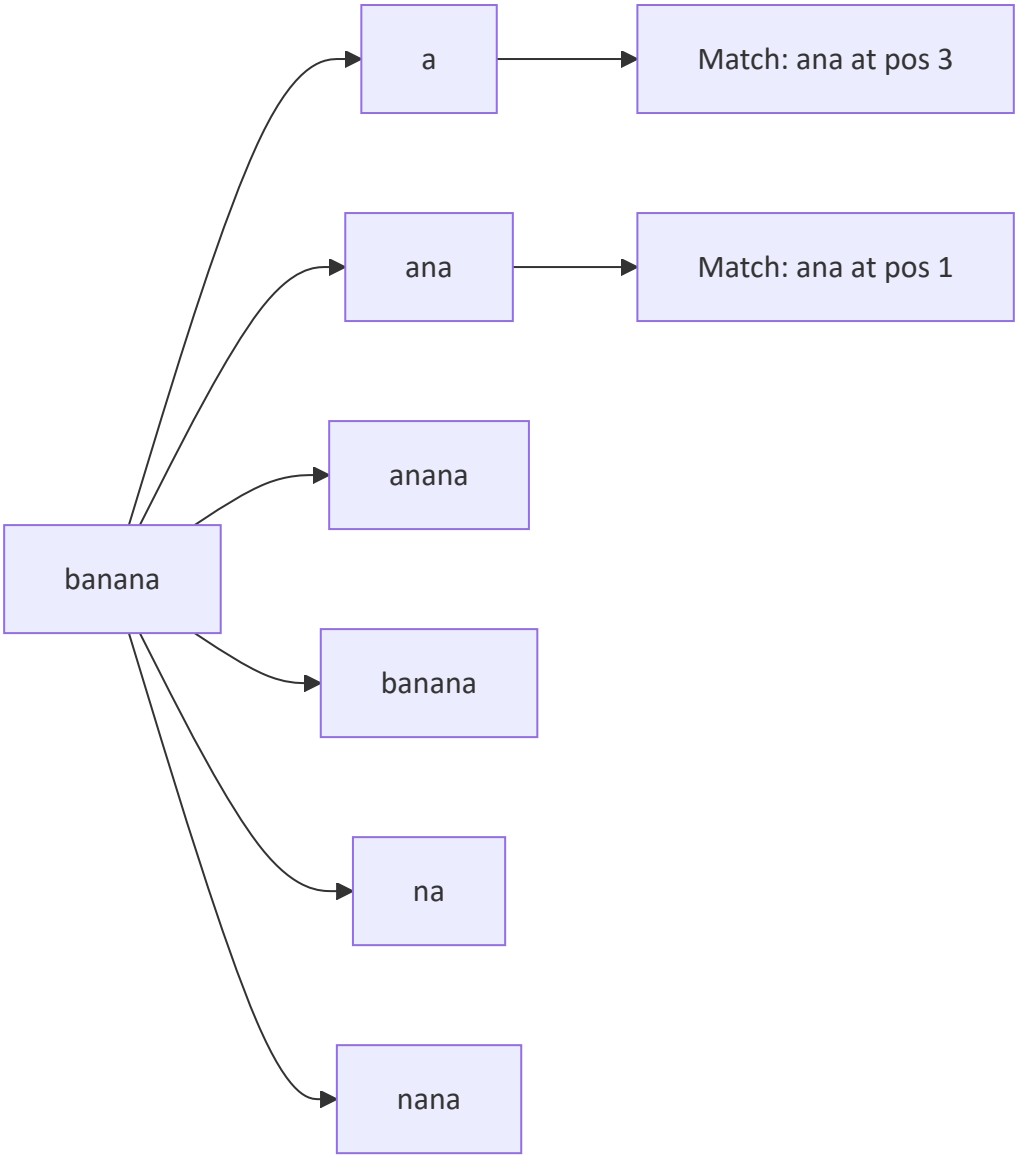
Rank	Suffix	Start Position
1	a	5
2	ana	3
3	anana	1
4	banana	0
5	na	4
6	nana	2

Suffix array: `[5, 3, 1, 0, 4, 2]`

Searching pattern 'ana':

- **Using suffix tree:** Traverse edges following characters `a → n → a`. The path exists and leads to leaves at positions 1 and 3.
- **Using suffix array:** Binary search for prefix `ana` among sorted suffixes. Matches found at ranks 2 and 3, corresponding to positions **3** and **1**.

Result: Pattern `ana` occurs in `banana` at positions **1** and **3**.



Explain query specification interfaces with examples (5 marks)

1. Boolean Interface

- Uses operators **AND, OR, NOT** to combine terms.
- **Example:** `(information AND retrieval) NOT data`
- **Strength:** Precise control; **Weakness:** Requires Boolean literacy.

2. Natural Language Interface

- Accepts plain-language questions or sentences.
- **Example:** "What is the difference between data retrieval and information retrieval?"
- **Strength:** Easy for casual users; **Weakness:** Ambiguous parsing.

3. Form-Based Interface

- Provides fields, dropdowns, and filters for structured input.
- **Example:** Google Advanced Search with fields for "all these words", "site", "file type".
- **Strength:** Guides users; **Weakness:** Can be rigid.

4. Graphical Interface

- Uses visual widgets such as sliders, Venn diagrams, or query builders.
- **Example:** A Venn diagram where users click regions to form Boolean queries.
- **Strength:** Intuitive for complex logic; **Weakness:** Screen-space intensive.

Result: Each interface trades off ease of use against expressive precision, and real systems often combine them.

Explain relevance judgments and relevance feedback in IR interfaces (5 marks)

Relevance Judgments: Assessments by users (or assessors) of whether a retrieved document satisfies their information need. They form the ground truth for evaluating precision, recall, and ranking quality.

Role in IR Interface:

- Users mark results as **relevant / not relevant**.
- The system uses these labels to refine the query or re-rank results.
- Enables personalized and adaptive search.

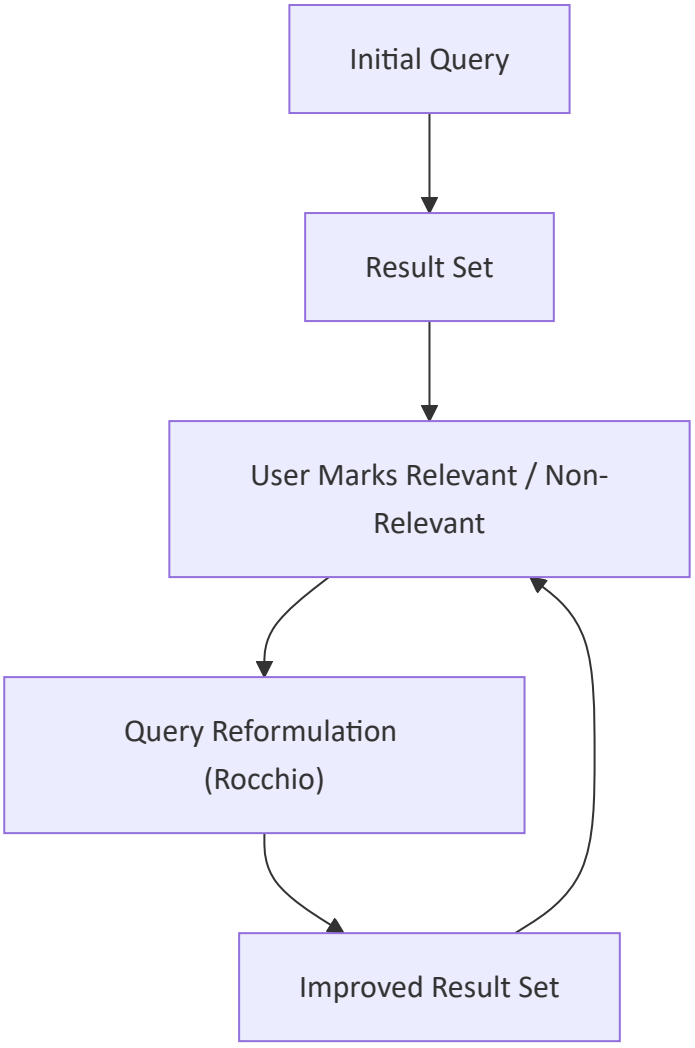
Relevance Feedback Process:

1. User issues an initial query.
2. System returns an initial result set.
3. User marks some results as relevant or non-relevant.
4. System reformulates the query (e.g., Rocchio algorithm) by expanding the query vector toward relevant documents and away from non-relevant ones.
5. Improved results are returned.

Rocchio Formula:

[
Q*{new} = \alpha Q + \beta \frac{1}{|D_r|}\sum*{d \in D_r} d - \gamma \frac{1}{|D_{nr}|}\sum*{d \in D_{nr}} d
]

- (D_r) = *relevant docs,* (D_{nr}) = non-relevant docs.
- (α, β, γ) control the balance.



Result: Relevance feedback closes the loop between user and system, progressively improving retrieval effectiveness.

Explain agglomerative hierarchical clustering; apply single-link on D1–D5 (5 marks)

Agglomerative Hierarchical Clustering: A bottom-up approach where each document starts as its own cluster, and the two closest clusters are repeatedly merged until one cluster remains.

Single-Link Method: Distance between two clusters = **minimum distance** between any pair of points across the clusters.

Given Distance Matrix:

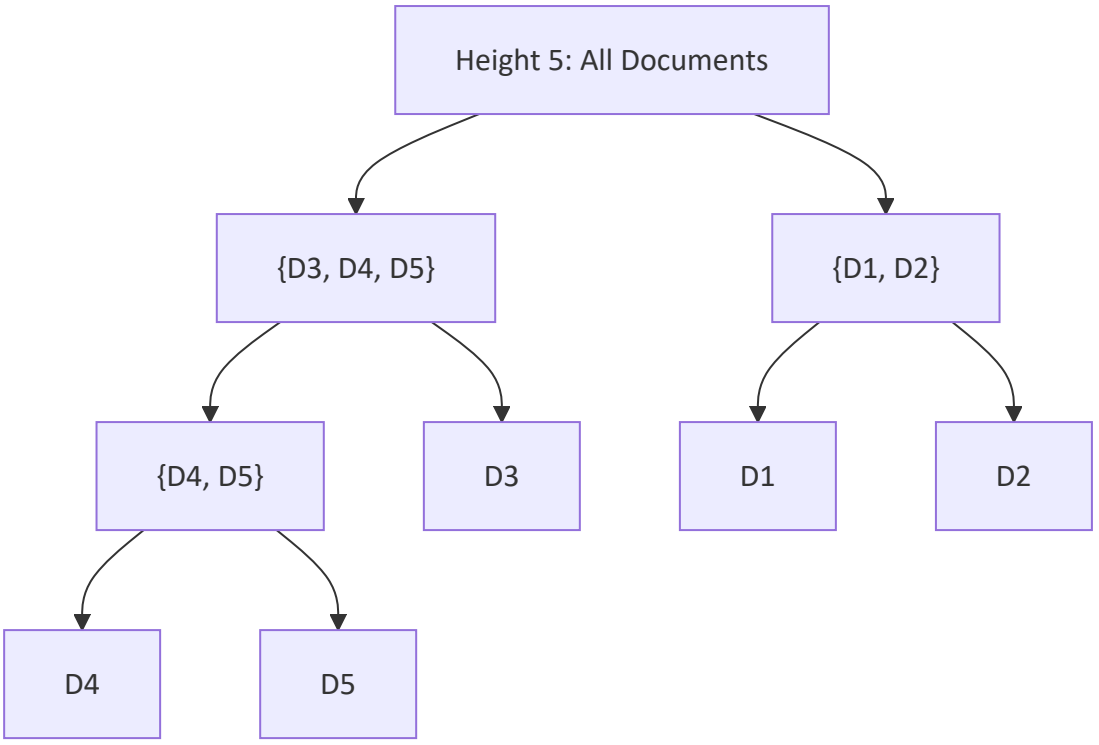
	D1	D2	D3	D4	D5
D1	0	2	6	10	9
D2	2	0	5	9	8
D3	6	5	0	4	5
D4	10	9	4	0	3
D5	9	8	5	3	0

Step-by-step merges:

1. **Smallest distance = 2** between D1 and D2 → merge **{D1, D2}** at height 2.
2. **Next smallest = 3** between D4 and D5 → merge **{D4, D5}** at height 3.

3. **Next smallest = 4** between D3 and D4 (part of {D4, D5}) → merge **{D3, D4, D5}** at height 4.
4. **Next smallest = 5** between {D1, D2} and {D3, D4, D5} (via D2–D3 = 5) → merge all at height 5.

Dendrogram:



Result: Merges occur at heights **2 (D1,D2), 3 (D4,D5), 4 (D3 with D4,D5), 5 (final)** — matching the answer check.

Construct inverted index and process Boolean queries (5 marks)

Documents (stop words removed: is, a, of, from, the):

- D1: information retrieval study information systems
- D2: web search type information retrieval
- D3: data retrieval different information retrieval

Term frequencies per document:

Term	D1	D2	D3
information	1	1	1
retrieval	1	1	1
study	1	0	0
systems	1	0	0
web	0	1	0
search	0	1	0
type	0	1	0
data	0	0	1
different	0	0	1

(a) Inverted Index (term, document id, term frequency):

Term	Postings (DocID, TF)
information	(D1, 1), (D2, 1), (D3, 1)
retrieval	(D1, 1), (D2, 1), (D3, 1)
study	(D1, 1)
systems	(D1, 1)
web	(D2, 1)
search	(D2, 1)
type	(D2, 1)
data	(D3, 1)
different	(D3, 1)

(b) Query Processing:

- **(information AND retrieval):** Intersection of postings → {D1, D2, D3}
- **AND NOT data:** Remove documents containing "data" (D3) → **{D1, D2}**
- **web OR data:** Union of postings → {D2} ∪ {D3} = **{D2, D3}**

Result: (information AND retrieval) AND NOT data = **{D1, D2}**; web OR data = **{D2, D3}** — matching the answer check.

Describe the stages of the Information Access Process with a diagram and evaluate interface support (5 marks)

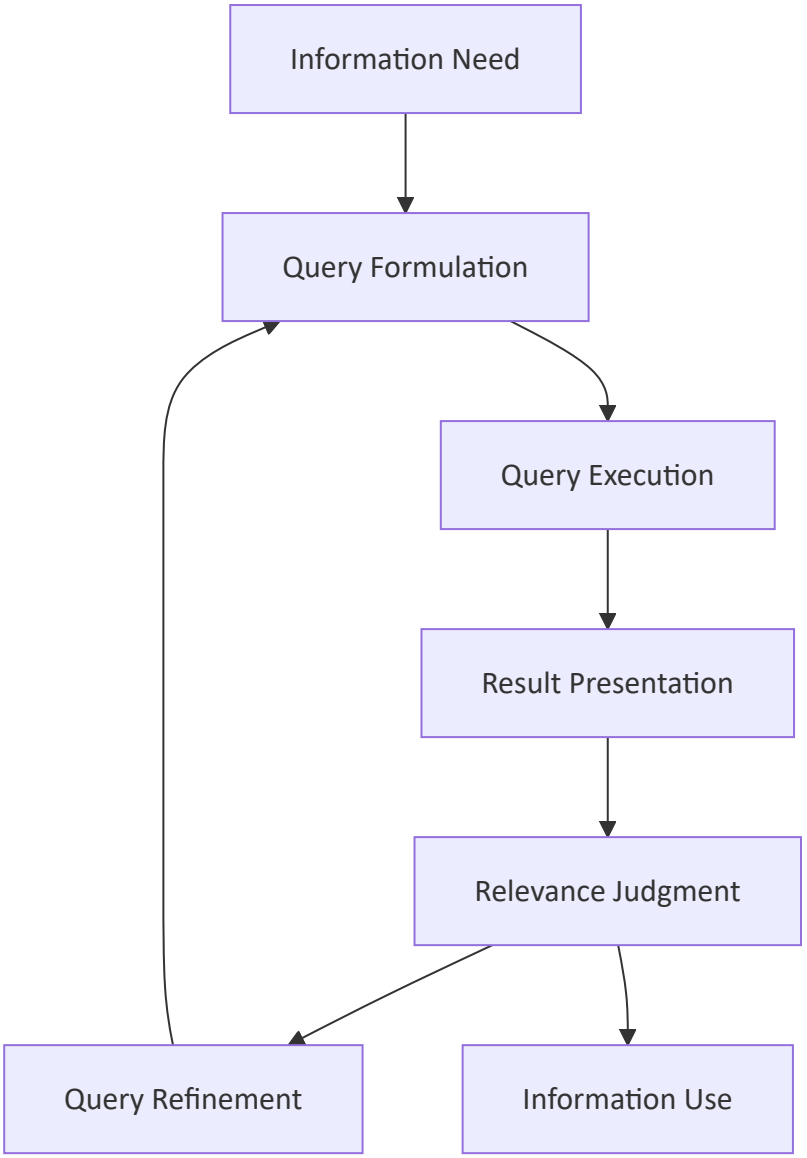
Stages of the Information Access Process:

1. **Information Need** — User recognizes a gap in knowledge.
2. **Query Formulation** — User translates the need into a query.
3. **Query Execution** — System matches the query against the index.
4. **Result Presentation** — Ranked results are displayed.
5. **Relevance Judgment** — User evaluates results.
6. **Query Refinement** — User reformulates based on feedback.
7. **Information Use** — User extracts and applies the information.

Interface Support at Each Stage:

Stage	Interface Support
Information Need	Starting points, suggested topics, browsing categories
Query Formulation	Auto-complete, Boolean builders, natural language input, form-based fields
Query Execution	Progress indicators, query previews
Result Presentation	Snippets, KWIC, clustering, visualization (SeeSoft, scatter plots)
Relevance Judgment	Relevance feedback buttons, highlighting, ranking explanations
Query Refinement	"More like this", query expansion, related searches

Stage	Interface Support
Information Use	Export, annotation, citation tools



Result: Interface support at each stage reduces cognitive load, improves query quality, and increases the likelihood of satisfying the user's information need.

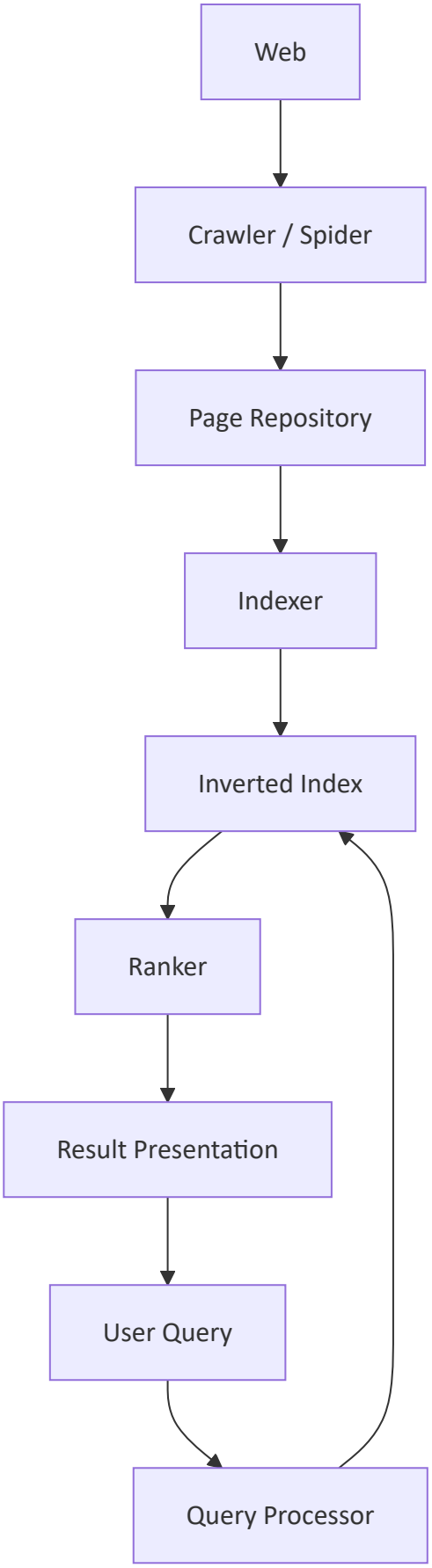
Analyze web search challenges and explain web search engine architecture (5 marks)

Challenges in Searching the Web:

- **Scale:** Billions of pages; distributed storage and parallel processing required.
- **Heterogeneity:** Diverse formats, languages, and quality levels.
- **Dynamic content:** Pages change, appear, and disappear rapidly.
- **Spam and SEO manipulation:** Adversarial content degrades ranking quality.
- **Deep web:** Content behind forms and databases is not directly crawlable.
- **Ambiguity:** Short queries with unclear intent.
- **Freshness:** Need for real-time indexing of news and social media.

Architecture of a Typical Web Search Engine:

1. **Crawler / Spider** — Traverses the web, fetches pages, and extracts links.
2. **Page Repository / Store** — Stores raw fetched pages.
3. **Indexer** — Parses pages, extracts terms, builds the inverted index.
4. **Index / Inverted Index** — Maps terms to document postings.
5. **Query Processor** — Parses the user query and retrieves candidate documents.
6. **Ranker** — Scores documents using link analysis (PageRank), relevance, and freshness.
7. **Result Presentation** — Displays ranked results with snippets and ads.



Result: A web search engine combines crawling, indexing, query processing, and ranking; its main challenges are scale, dynamism, heterogeneity, and adversarial content.